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SOLVED NUMERICALS – JAN/FEB 2025 (CSE STREAM)

1(C) Given the numerical aperture 0.31 and refractive index of core is 1.49. Calculate the critical angle for the core-cladding interface. **(Jan-2025-22S-CSE STREAM)**

Given data: Numerical Aperture (NA) = 0.31and refractive index of core $(n_1) = 1.49$

$$NA = \sqrt{(n_1^2 - n_2^2)}$$

Rearranging, $n_2 = \sqrt{(n_1^2 - NA^2)} = \sqrt{(1.49^2 - 0.31^2)} = 1.46$

$$\theta_c = sin^{-1}\left(\frac{n_2}{n_1}\right) = sin^{-1}\left(\frac{1.46}{1.49}\right) = 78°28°$$

2(C) In an optical fiber experiment the LASER light propagating through optical fibre cable is 1.5m, made a spot diameter of 8mm on the screen. The distance between the end of the optical fibre cable and the screen is 0.031m. Calculate angle of contact and NA of given optical fibre. **(Jan-2025-22S-CSE STREAM)**

Given data: Diameter, D = 8mm and L = 0.031m

$$\boldsymbol{\theta = tan^{-1}\left(\frac{D}{2L}\right) = tan^{-1}\left(\frac{8\times10^{-3}}{2\times0.031}\right) = 7.35°}$$

3(C) Calculate the first three energy states for an electron in one dimensional potential well of width of 0.1nm. **(Jan-2025-22S-CSE STREAM)**

Data: *Width of the potential well,* $a = 0.1nm = 0.1 \times 10^{-9}m$

Tofind:

Values of energy of the electron in the ground and the first two excited states

i.e $E_1 = ?$ *i.e., Ground state,* $E_2 = ?$ *(First excited state),* $E_3 = ?$ *(Second excited state)*

Solution: The energy of the electron in a one dimensional infinite potential well is given by,

$$E_n = \frac{n^2h^2}{8ma^2}$$

$$\therefore \text{First energy state,} \quad E_1 = \frac{n^2h^2}{8ma^2} = \frac{1^2(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(0.1^{-9})^2}$$

$$\therefore E_0 = E_1 = 6.0314\times10^{-18}J = 37.64\ eV$$

First Excited State Energy n=2 , $E_2 = 4E_1 = 24.115\times10^{-18}J = 150.72\,eV$

First Excited State Energy n=2 , $E_3 = 9E_1 = 54.260\times10^{-18}J = 339.12\,eV$

4(C) The kinetic energy of an electron is equal to energy of photon with a wavelength of 560nm. Calculate the de Broglie wavelength of the electron. **(Jan-2025-22S-CSE STREAM)**

The kinetic energy of an electron = energy of photon with $\lambda = 560\times10^{-9}m$

$$E = \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times3\times10^{8}}{560\times10^{-9}} = 3.549\times10^{-19}J$$

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we have the equation for de broglie wavelength $, \lambda = \frac{h}{\sqrt{2mE}}$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 3.549 \times 10^{-19}}} = 8.24 \times 10^{-10} m$$

Given $|\Psi> = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$ and $|\phi> = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$, Prove that $<\Psi|\phi> = <\phi|\Psi>^*$.

5(C)

Given $|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$ and $|\phi\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$ Prove that $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$ Arriving at (1)

Solution : $\langle\phi|\psi\rangle = \begin{pmatrix} \beta_1^* & \beta_2^* \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \beta_1^*\alpha_1 + \beta_2^*\alpha_2$.....(1)

$\langle\psi|\phi\rangle = \begin{pmatrix} \alpha_1^* & \alpha_2^* \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \beta_1\alpha_1^* + \beta_2\alpha_2^*$ Arriving at (2)

$\langle\psi|\phi\rangle^* = (\beta_1\alpha_1^* + \beta_2\alpha_2^*)^* = \beta_1^*\alpha_1 + \beta_2^*\alpha_2$.....(2)

Thus from (1) and (2) Comparing the (1) and (2)

$\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$

6(C) *A linear operator 'X' operates such that* $X|0\rangle = |1\rangle$ *and* $X|1\rangle = |0\rangle$. Find the matrix representation of 'X'. **(Jan-2025-22S-CSE STREAM)**

Solution: $\begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}$, $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ *and* $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Given $X|0\rangle = |1\rangle$ $\quad X|0\rangle = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Multiply the matrices on LHS and equating with the matrix on the RHS we get,

$x_{11} = 0$ & $x_{21} = 1$ *Given* $X|1\rangle = |0\rangle$ $\quad X|1\rangle = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Multiply the matrices on LHS and equating with the matrix on the RHS we get,

$x_{12} = 1$ & $x_{22} = 0$ *Therefore,* $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

7(C) Calculate the probability of an electron occupying an energy level 0.2eV above the Fermi level at 27°C.

Data: $E - E_F = 0.2\text{eV} = 0.2 \times 1.602 \times 10^{-19} J$

To find: $f(E)$ *at* $27°C$ *(i.e need to convert into kelvin is* $300K$)

(i)Evaluation of $f(E)$ *at* $300K = ?$

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1} = \frac{1}{e^{\frac{0.2 \times 1.602 \times 10^{-19}}{1.38 \times 10^{-23} \times 300}} + 1} = \frac{1}{e^{7.7891} + 1} = \frac{1}{2296.4 + 1} = \frac{1}{2297.4} = 4.35 \times 10^{-4}$$

$\therefore f(E)$ *at* $300K = 4.35 \times 10^{-4}$

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8(C) A superconducting tin has a critical temperature of 3.7 K at zero magnetic field and a critical field of 0.0306 Tesla at 0 K. Find the critical field at 2 K.

(Jan-2025-22S-CSE STREAM)

Solution:

Given data:

Critical temperature $T_C = 3.7$ K

Critical field $H_C = 0.0306$ Tesla

Temperature $T = 2$ K

The critical magnetic field

$$H_C = H_0\left[1-\left[\frac{T^2}{T_c^2}\right]\right]$$

$$= 0.0306\left[1-\left[\frac{2}{3.7}\right]^2\right] = 0.0216 \text{ Tesla}$$

The critical magnetic field $H_C = 0.0216$ Tesla.

9(C). A slowing in object in an animation has a first frame distance 0.5m and first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.

(Jan-2025-22S-CSE STREAM)

$$base\ distance = \frac{distance\ between\ any\ two\ adjacent\ frames}{2}$$

$$base\ distance = \frac{0.5-0.35}{2} = 0.075m$$

To find number of frames $no\ of\ frames = \frac{total\ distance}{base\ distance}$

$$No\ of\ frames = \frac{0.5}{0.075} = 6.66 \approx 7\ frames$$

10(C) In a laser diffraction grating experiment the laser light undergoes second order diffraction for the diffraction angle 1.48°. The grating constant $d = 5.05 \times 10^{-5} m$ and the distance between the grating and screen is 0.60m. Find the wavelength of the LASER light.

(Jan-2025-22S-CSE STREAM)

Formula $\lambda = \frac{d\sin(\theta)}{n}$ n=2,

Substitution $\lambda = \frac{5.05\times10^{-5}\sin(1.48^o)}{2} m$

Calculation and Final Answer $\lambda = 6.522\text{x}10^{-7}$ m

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